MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) – MG4SECMBG201 – FUNDAMENTALS OF MICROBIOLOGICAL ANALYSIS OF FOOD AND WATER

Part A – Multiple Choice Questions – Answer *all* questions – Each question carries *1 mark*

1. (a) toxins
2. (c) pink
3. (b) Class II or Class III BSC
4. (c) Acetone treatment
5. (a) Colony forming unit
6. (b) blue to white
7. (c) yellow
8. (b) mixed populations
9. (a) 0.45 μm
10. (c) Lactose

Part B – True or False questions – Answer *all* questions – Each question carries *1 mark*

11. False
12. False
13. True
14. True
15. True

Part C – One word or very short answer questions – Answer *all* questions – Each question carries *1 mark*

16. Sterile containers made of glass or plastic
17. Prevent exposure to microbes
18. *E. coli* produce pink to red colonies; enterohemorrhagic *E. coli* O157:H7 produce colourless or pale colonies on SMAC agar
19. To prevents leakage of fumigant gas and to ensure the effective action of the fumigant gas in required concentrations
20. MacConkey broth/Brilliant Green Lactose Bile (BGLB) broth (any one, 1 mark)
21. 25
22. Indicator organisms are microorganisms whose presence in a sample indicates possible contamination or the potential presence of pathogenic microorganisms. eg. *E. coli* indicates faecal contamination and also

possible presence of intestinal pathogens like *Salmonella*. (either correct definition or *E. coli* as example – 1 mark)

23. One nation one food law
24. L- shaped / T shaped glass rod (any one)
25. Inoculation loop

Part D - Reason based short answer questions - Answer *any 5* questions - Each question carries *1 mark*

26. To ensure that all unwanted microorganisms that may already be present in the medium are destroyed before inoculating the desired organism.
27. Stains add colour to the cells, creating contrast between the bacteria and the background, which makes them easier to see.
28. Baird-Parker Agar is selective because it suppresses many non-staphylococcal bacteria and differential because it allows differentiation of coagulase-positive staphylococci from coagulase-negative staphylococci. (selective – ½ mark, differential – ½ mark)
29. The nutrient medium itself may absorb some light**;** using it as a blank eliminates this background absorbance – Helps to ensure that the absorbance measured is only due to the sample and not the medium. (any one, 1 mark)
30. Proper rotation and rolling increase the chances of collecting representative microorganisms.
31. Keeping only essential materials in the working area helps maintain a clean, safe, and contamination-free laboratory environment.
32. SS agar is a selective and differential media**.** It inhibits the growth of many Gram-positive bacteria and some Gram-negative intestinal flora, but allows the growth of enteric pathogens such as *Salmonella* and *Shigella*. The colony characters differentiate the growth of *Salmonella*, *Shigella* and *E. coli*. (selective – ½ mark, differential – ½ mark)
33. Exposure to air increases oxidation of fats, leading to rapid spoilage and rancidity.

Part E – Short answer questions – Answer *any 5* questions – Each question carries *2 marks*

34. Explain chemical hazards – Examples of chemical hazards and importance in food safety (1 mark each)
35. Procedure of acid value estimation in brief.
36. Significance of EU standards – Role in ensuring food safety, food quality, consumer confidence, promotion of internationa trade etc (any 2 relevant points, 1 mark each)
37. Definition of inoculum and inocuation (1 mark each)
38. Components of PCA: Tryptone, Beef extract, Dextrose, Agar, Distilled water (1 mark) – Preparation procedure (1 mark)
39. Role of lactic acid, phenol, cotton blue and glycerol in LPCB in the staining of fungi (1 mark) – Use of LPCB in the study of fungal morphology and its identification. (1 mark)

40. Brief description of Presumptive Test, its procedure (1 mark) – Its role in the presumptive identification of coliforms in water (1 mark).
41. Composition of EMB agar: Peptone, lactose, dipotassium hydrogen phosphate, eosin Y, methylene blue, agar agar and distilled water.

Part F – Long answer questions – Answer *any 2* questions – Each question carries *5 marks*

42. Describe on sampling and its significance in testing (2 marks) – Example of industrial and pharmaceutical samples (1 mark) – Used for quality control and sterility testing (2 marks).
43. Sterilization using dry heat with hot air oven as example (2½ marks) – Sterilization using dry heat with autoclave as example (2½ marks).
44. Mannitol Salt Agar is a selective and differential medium used for the isolation and differentiation of *Staphylococcus* species, particularly *Staphylococcus aureus* (1 mark) – Growth characteristics of mannitol fermenters like *S. aureus* and non-fermenters on MSA medium (2 marks each).
45. Definition and principle of acid value (2½ marks) – Definition and principle of peroxide value (2½ marks).